Akoizo Original Practice - Form 4

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What gas inflates the balloon in the vinegar and baking soda action?

A. Hydrogen

B. Nitrogen

C. Carbon dioxide

D. Oxygen

Answer: _____

2. (2 points) Which property most cleanly distinguishes a quarter from a nickel?

A. Color

B. Magnetism

C. Diameter

D. Mass

Answer: _____

3. (2 points) Action 6 starts before action 5 on the ASL. What happens to action 5?

A. It earns zero points, even if it completes later

B. It still earns 50

C. It earns 25

D. The run restarts

Answer: _____

4. (2 points) Which component is NOT allowed?

Unit: Electrical, infrared, and magnetic actions.
Worked-case data: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?
Separate investigation observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

A. A microcontroller

B. A relay

C. An IR LED

D. A 9 V battery

Answer: _____

5. (4 points) With a pulley IMA of 3, how far must the effort rope move to lift a load 12 cm?

Answer: _____
_____
_____

6. (4 points) Calculate the size points for a device measuring 70.0 x 45.0 x 79.5 cm.

Answer: _____
_____
_____

7. (4 points) Why aim slightly under the Target Time?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

Answer: _____
_____
_____

8. (4 points) Why should reactants be pre-measured and labeled?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

Answer: _____
_____
_____

9. (4 points) Why should the first action be clearly visible?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

Answer: _____
_____
_____

10. (4 points) How can a team earn the 75-point setup bonus at State?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?

Answer: _____
_____
_____

12. (2 points) Does the pulley action qualify?

Supplied synthetic observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

A. Yes; it lifts more than 10 cm

B. No; IMA is only 2

C. Yes; the load exceeds 400 g

Answer: _____

13. (2 points) Define “Start Action” in the context of Mission Possible.

Answer: _____
_____
_____

14. (2 points) Define “Target Operation Time” in the context of Mission Possible.

Answer: _____
_____
_____

15. (2 points) Define “Acetic acid” in the context of Mission Possible.

Answer: _____
_____
_____

16. (2 points) Define “Effort arm” in the context of Mission Possible.

Answer: _____
_____
_____

17. (2 points) Define “First action” in the context of Mission Possible.

Answer: _____
_____
_____

18. (2 points) Define “Dead-end action” in the context of Mission Possible.

Answer: _____
_____
_____